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3D PRINTED FURNITURE WITH CAPTURED COMPONENTS AND METHOD OF MAKING THE SAME

Abstract

A three-dimensional (3D) printed multi-component structure includes a first component and a second component. The second component is formed to define a cavity that extends into the second component. The first component is located in the cavity of the second component to at least partially embed the first component in the second component.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to and the benefit of U.S. Provisional Patent Application Number 63/558,725, filed 28 Feb. 2024, the disclosure of which is now expressly incorporated herein by reference.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates generally to furniture, and more specifically to 3D printed furniture with captured components and methods of the making the same.

BACKGROUND

[0003] Disposing of some objects, such as furniture or décor, may be difficult. Furniture waste is becoming increasingly problematic as furniture is generally made of multiple different materials, many of which are difficult to recycle. Further, taking apart furniture to process each type of material separately is time-consuming and expensive.

[0004] Three-dimensional (3D) printing may be used to make three-dimensional objects. 3D printing is achieved using additive processes during which an object is created by laying down successive layers of material until the object is created. There remains interest in using 3D printing to reduce furniture waste.

SUMMARY

[0005] The present disclosure may comprise one or more of the following features and combinations thereof.

[0006] A method of three-dimensional (3D) printing multi-component furniture may comprise providing a first furniture component. The method may comprise depositing at least one first layer of molten material on a print surface to form a first side of a second furniture component. The method may comprise depositing at least one second layer of molten material onto the at least one first layer of molten material while the at least one first layer of molten material is molten to form a body of the second furniture component having a cavity. The method may comprise inserting the first furniture component into the cavity of the second furniture component. The method may comprise depositing at least one third layer of molten material onto the at least one second layer of molten material and the first furniture component while the at least one second layer of molten material is molten to form a second side of the second furniture component. The method may comprise cooling the at least one first layer of molten material, the at least one second layer of molten material, and the at least one third layer of molten material to cause the at least one first layer of molten material, the at least one second layer of molten material, and the at least one third layer of molten material to solidify and form the second furniture component having the first furniture component at least partially encapsulated therein.

[0007] In some embodiments, depositing at least one third layer of molten material onto the at least one second layer of molten material and the first furniture component may include depositing the at least one third layer of molten material on a perimeter of the first furniture component so that the first furniture component is captured in the second furniture component by the at least one third layer of molten material and at least a portion of the first furniture component is accessible from an exterior of the second furniture component. Depositing at least one third layer of molten material onto the at least one second layer of molten material and the first furniture component may include depositing the at least one third layer of molten material on an entirety of the first furniture component so that the first furniture component is entirely encapsulated in the second furniture

component by the at least one third layer of molten material.

[0008] In some embodiments, the first furniture component may not be accessible from an exterior of the second furniture component. The method may further comprise, after the step of cooling, removing material from the second side of the second furniture component located above the first furniture component to form an opening in the second side of the second furniture component so that the first furniture component is accessible from an exterior of the second furniture component. Removing material from the second side of the second furniture component may include milling the material from the second side of the second furniture component.

[0009] In some embodiments, the method may further comprise providing a metal plate. The method may further comprise, after the step of depositing at least one first layer of molten material on a print surface and before the step of depositing at least one second layer of molten material onto the at least one first layer of molten material, locating the metal plate on the at least one first layer of molten material and depositing at least one fourth layer of molten material on the at least one first layer of molten material and the metal plate. The method may further comprise, after the step of cooling, heating the metal plate to cause the solidified at least one third layer of molten material and the solidified at least one fourth layer of molten material located on the metal plate to become molten.

[0010] In some embodiments, the method may further comprise removing the metal plate and the first furniture component from the second furniture component. The first furniture component may include an RFID chip or a NFC chip. The first furniture component may include a sensor. The method may further comprise 3D printing the first furniture component.

[0011] According to another aspect of the present disclosure, a method of three-dimensional (3D) printing a multi-component structure may comprise providing a first component. The method may comprise depositing at least one first layer of molten material to form a first side of a second component. The method may comprise depositing at least one second layer of molten material onto the at least one first layer of molten material while the at least one first layer of molten material is molten to form a body of the second component. The method may comprise positioning the first component on the second component. The method may comprise depositing at least one third layer of molten material onto the first component while the at least one second layer of molten material is molten to form a second side of the second component. The method may comprise cooling the at least one first layer of molten material, the at least one second layer of molten material, and the at least one third layer of molten material to cause the at least one first layer of molten material, the at least one second layer of molten material, and the at least one third layer of molten material to solidify and form the second component having the first component at least partially embedded therein.

[0012] In some embodiments, positioning the first component on the second component may include positioning an embedded portion of the first component on top of the body of the second component and positioning an external portion of the first component adjacent to the body of the second component. Depositing at least one third layer of molten material onto the first component may include directing the at least one third layer of molten material through at least one hole formed in the embedded portion of the first component. The external portion of the first component may be formed to include a plurality of ridges that align with the first layer of molten material and the second layer of molten material.

[0013] According to another aspect of the present disclosure, a three-dimensional (3D) printed multi-component structure may comprise a first component and a second component. The second component may have a first side and a second side opposite the first side. The second component may be formed to define a cavity that extends into the second component from the second side toward the first side. The second component may be 3D printed and the first component may be located in the cavity of the second component to at least partially embed the first component in the second component.

[0014] In some embodiments, the first component may be entirely encapsulated in the second component. The first component may comprise a RFID chip, a NFC chip, or a sensor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The detailed description particularly refers to the accompanying figures in which:

[0016] FIG. 1 is a diagrammatic view of a method for three-dimensional (3D) printing a multi-component structure showing that the method includes (i) providing a first furniture component, (ii) setting a print path for a first layer of molten material and depositing the first layer to form a first side of a second furniture component, (iii) setting a print path for a second layer of molten material and depositing the second layer on the first layer, (iv) setting a print path for a third layer of molten material and depositing the third layer on the second layer to form a cavity in the third layer of molten material, (v) inserting the first furniture component into the cavity, and (vi) depositing a fourth layer of molten material on an entirety of the first furniture component and the third layer of molten material to form the second furniture component having the first furniture component encapsulated therein;

[0017] FIG. 2 is a diagrammatic view of a method for three-dimensional (3D) printing the multi-component structure showing that the method includes milling an opening in the second furniture component so that the first furniture component is visible;

[0018] FIG. 3 is a diagrammatic view of a method for three-dimensional (3D) printing the multi-component structure showing that the method includes providing a metal plate and locating the metal plate on the first layer of molten material so that the metal plate is encapsulated by the second layer of molten material;

[0019] FIG. 4 is a diagrammatic view of an alternative method for three-dimensional (3D) printing a multi-component structure showing that the method includes (i) providing a first furniture component, (ii) setting a print path for a first layer of molten material and depositing the first layer to form a first side of a second furniture component, (iii) setting a print path for a second layer of molten material and depositing the second layer on the first layer, (iv) setting a print path for a third layer of molten material and depositing the third layer of molten material on the second layer to form a cavity in the third layer, (v) inserting the first furniture component into the cavity, and (vi) depositing a fourth layer of molten material on a perimeter of the first furniture component and on the third layer of molten material to form the second furniture component having the first furniture component captured therein;

[0020] FIG. 5 is a diagrammatic view of an alternative method for three-dimensional (3D) printing a multi-component structure showing that the method includes (i) providing a first furniture component, (ii) setting a print path for a first layer of molten material and depositing the first layer to form a first side of a second furniture component, (iii) locating the first furniture component on the first layer of molten material, and (vi) depositing a second layer of molten material on a perimeter of the first furniture component and on the first layer of molten material to form the second furniture component having the first furniture component captured therein; and

[0021] FIG. 6 is a diagrammatic view of an alternative method for three-dimensional (3D) printing a multi-component structure showing that the method includes (i) providing a first furniture component, (ii) depositing a first layer of molten material to form a first side of a second furniture component and depositing a second layer of molten material on the first layer to form a body of the second furniture component, (iii) locating a portion of the first furniture component on the body of the second furniture component, and (iv) depositing a third layer of molten material on the first furniture component to form the second furniture component having the first furniture component partially embedded therein.

DETAILED DESCRIPTION

[0022] For the purposes of promoting an understanding of the principles of the disclosure, reference will now be made to a number of illustrative embodiments illustrated in the drawings and specific language will be used to describe the same.

[0023] A three-dimensional (3D) printed multi-component structure **10** in accordance with the present disclosure includes a first furniture component **12** and a second furniture component **14**, as shown in FIGS. **1-3**. The first furniture component **12** may be captured by or embedded in the second furniture component **14** without using adhesives and/or fasteners, such as glue, bolts, screws, rivets, nails, or other similar adhesives or fasteners. Instead, the first furniture component **12** is inserted into the second furniture component **14** while the second furniture component **14** is being 3D printed so that the first furniture component **12** is captured by or embedded in the second furniture component **14** during the printing process. The absence of adhesives and/or fasteners allows for direct recycling of the second furniture component **14** after removal of the first furniture component **12**, for example, at an end of life of the multi-component structure **10**.

[0024] The first furniture component **12** may provide functionality and/or aesthetic appeal to the second furniture component **14**. In illustrative embodiments, the first furniture component **12** is an object or a plurality of objects that is captured by or embedded in the second furniture component **14**. In some embodiments, the first furniture component **12** is 3D printed. In some embodiments, the first furniture component **12** is not 3D printed.

[0025] The second furniture component **14** includes a first side **16**, a second side **18** opposite the first side **16**, and a body **17** extending between the first side **16** and the second side **18**, as shown in FIG. **1**. The first furniture component **12** is located between the first side **16** and the second side **18** of the second furniture component **14**. The second furniture component **14** is formed to define a cavity **20** that extends into the body **17** toward the first side **16**. The first furniture component **12** is positioned in the cavity **20**.

[0026] In illustrative embodiments, the second furniture component **14** is 3D printed. The second furniture component **14** may be formed as, for example, a table, a table base, a bench, a chair, a couch, a planter, a desk, a cubicle, a wall or table decoration, or a frame, among other furniture items.

[0027] The first furniture component **12** may comprise an active and/or interactive component configured to return feedback to a user of the multi-component structure **10** or to interact with the user of the multi-component structure **10**. The first furniture component **12** may be an electronic component. For example, the first furniture component **12** may comprise an NFC chip or an RFID chip that transmits information to the user while the user is in close proximity to the multi-component structure **10**. The NFC chip or the RFID chip may be configured to direct the user to a website or an application. The information transmitted to the user may include information such as the history of the material used to create the second furniture component **14**, the designer and/or the engineer of the second furniture component **14**, the inspiration behind the second furniture component **14**, or data regarding other users of the second furniture component **14**, among other information.

[0028] As another example, the first furniture component **12** may comprise a touch screen device such as a tablet or a remote control for TV, fans, lights, audio systems, or other appliances. As another example, the first furniture component **12** may comprise a remote control including buttons to control TV, fans, lights, audio systems, or other appliances.

[0029] The first furniture component **12** may comprise a functional, non-interactive component configured to collect data and/or perform a function for the user of the multi-component structure **10**. For example, the first furniture component **12** may comprise a sensor configured to collect and store data. The sensor may comprise a temperature sensor, a proximity sensor, a humidity sensor, a pressure sensor, a force sensor, or any other suitable sensor or combination of sensors. The sensor may collect and store data regarding interactions with the multi-component structure **10**, such as,

but not limited to, time stamps of when users are interacting with the multi-component structure **10**. For example, the sensor may store time stamp data every time a user sits on a bench (i.e., the second furniture component **14**).

[0030] As an example, the sensor may collect and store weather data, such as temperature data and humidity data. As another example, the sensor may collect and store data related to the health and/or the status of the multi-component structure **10**. As an additional example, the sensor may collect and store data related to a location of the multi-component structure **10**. The sensor may store the information locally so that the information may be retrieved via a memory card, or the sensor may transmit the information to a remote storage network.

[0031] As another example of functional, non-interactive components, the first furniture component **12** may include bearings, hinges, drawers, or other similar components. The first furniture component **12** may include a drain in a planter (i.e., the second furniture component **14**). The first furniture component **12** may include a heater in a bench, a chair, or a couch (i.e., the second furniture component **14**) to warm the bench, the chair, or the couch. The first furniture component **12** may include a glass window in a window frame (i.e., the second furniture component **14**), as shown in FIG. 5. The first furniture component **12** may include a speaker in a speaker box (i.e., the second furniture component **14**). The first furniture component **12** may include controls for a motor in a recliner (i.e., the second furniture component **14**). The first furniture component **12** may include wiring and/or LEDs for the second furniture component **14**. The first furniture component **12** may include one or more USB ports for charging electronics. The first furniture component **12** may include hooks for coats, purses, bags, keys, umbrellas, etc. The first furniture component **12** may include metal mesh to create a stronger second furniture component **14**. The first furniture component **12** may include a holder, such as a cup holder.

[0032] The first furniture component **12** may comprise passive components configured to add aesthetic appeal to the second furniture component **14**. For example, the first furniture component **12** may include a placard or a sign, as shown in FIG. 6. As another example, the first furniture component **12** may include a table top, such as a wooden top, a marble top, a metal top, a non-printed polymeric material based top, a ceramic top, a stone top, an engineered materials based top, a polymer top, etc., for a table base (i.e., the second furniture component **14**). The first furniture component **12** may include an inlay formed of any material that provides color and/or material contrast from the second furniture component **14**.

[0033] A method **200** of three-dimensional (3D) printing the multi-component structure **10** is shown in FIGS. 1-3. In step **202**, the first furniture component **12** is provided. In step **204**, a print path for a nozzle of a printer to deposit at least one first layer of molten material on a print surface is set. In step **206**, the at least one first layer of molten material is deposited on the print surface following the print path. The at least one first layer of molten material forms the first side **16** of the second furniture component **14**. In step **208**, a print path for the nozzle to deposit at least one second layer of molten material on the at least one first layer is set. In some embodiments, the print path for the at least one second layer of molten material is different than the print path for the at least one first layer of molten material, as suggested in FIG. 1.

[0034] In step **210**, the at least one second layer of molten material is deposited on the at least one first layer of molten material while the at least one first layer of molten material is molten. The at least one second layer of molten material forms a portion of the body **17** of the second furniture component **14**.

[0035] In step **212**, a print path for the nozzle to deposit at least one third layer of molten material on the at least one second layer is set, as shown in FIG. 1. The print path for the at least one third layer of molten material may be different than the print path for the at least one first layer of molten material and/or the print path for the at least one second layer of molten material. The print path for the nozzle in step **212** is based, at least in part, on a shape of the first furniture component **12**.

[0036] In step **214**, the at least one third layer of molten material is deposited on the at least one

second layer while the at least one second layer of molten material is molten. The at least one third layer of molten material forms another portion of the body **17** of the second furniture component **14**. The print path of the at least one third layer forms the cavity **20** in the body **17** of the second furniture component **14** as the at least one second layer of molten material is not entirely covered by the at least one third layer, as shown in FIG. **1**. The cavity **20** may have any shape that allows the first furniture component **12** to be positioned in the cavity **20**. It will be understood that any number of layers of molten material may be deposited on top of one another before the cavity **20** is formed, depending, for example, on the desired shape of the second furniture component **14**.

[0037] In step **216**, the first furniture component **12** is inserted into the cavity **20** of the second furniture component **14**, as shown in FIG. **1**. In step **218**, a print path for the nozzle to deposit at least one fourth layer of molten material is set. In step **220**, the at least one fourth layer of molten material is deposited onto the at least one third layer of molten material and the first furniture component **12** while the at least one third layer of molten material is molten to form the second side **18** of the second furniture component **14**. The at least one fourth layer of molten material bonds to the at least one third layer of molten material to hold the first furniture component **12** in place in the cavity **20**. In some embodiments, the at least one fourth layer of molten material covers an entirety of the first furniture component **12** such that the first furniture component **12** is encapsulated in the second furniture component **14** and not visible from an exterior of the second furniture component **14**, as shown in FIG. **1**.

[0038] Any number of layers of molten material may be deposited on top of the first furniture component **12**. The number of layers is dependent, at least in part, on the type of the first furniture component **12** and the second furniture component **14**. For example, if the first furniture component **12** is an RFID chip, too many layers deposited on top of the RFID chip may interrupt the functionality of the RFID chip.

[0039] In step **222**, the layers of molten material are cooled to cause the layers of molten material to solidify and form the second furniture component **14** having the first furniture component **12** encapsulated therein, as shown in FIG. **1**. As the at least one third layer of molten material cools, the at least one third layer of molten material may shrink around the first furniture component **12** to act as a clamp to hold the first furniture component **12** in place. The at least one second layer and the at least one fourth layer may also act as a clamp to hold the first furniture component **12** in place.

[0040] In some embodiments, method **200** further includes step **224** of removing material from the second side **18** of the second furniture component **14** located above the first furniture component **12** to form an opening **22** in the second side **18** of the second furniture component **14**, as shown in FIG. **2**. The opening **22** provides visibility and/or accessibility of and to the first furniture component **12** from an exterior of the second furniture component **14**. In some embodiments, removing the material from the second side **18** of the second furniture component **14** includes milling the material from the second side **18** of the second furniture component **14**. In some embodiments, removing the material from the second side **18** of the second furniture component **14** includes turning, drilling, grinding, cutting, lasering, or any other subtractive process.

[0041] The opening **22** may be formed if the first furniture component **12** is an electrical component that may require connection via a cable for data transfer. As another example, the opening **22** may be formed if the first furniture component **12** is an inlay and a smooth surface in plane with the inlay is desired. The opening **22** may have any shape such as, for example, a rectangular shape, a circular shape, an irregular shape, among others.

[0042] In some embodiments, to recycle the multi-component structure **10**, the portion of the second furniture component **14** surrounding the first furniture component **12** is removed so that the first furniture component **12** may be extracted from the second furniture component **14**. Removal of the first furniture component **12** from the second furniture component **14** allows for separate discarding of the non-recyclable or dissimilar recyclable material of the first furniture component

12.

[0043] In some embodiments, for recycling and/or removal purposes, the portion of the second furniture component **14** surrounding the first furniture component **12** is removed via milling, turning, drilling, grinding, cutting, lasering, or any other subtractive process. In some embodiments, for recycling and/or removal purposes, the portion of the second furniture component **14** surrounding the first furniture component **12** is removed via increasing the temperature of the portion of the second furniture component **14**.

[0044] In some embodiments, as shown in FIG. 3, step **202** of method **200** may include providing a metal plate **24**. The metal plate **24** may comprise ferrous metal. In step **206**, the metal plate **24** may be positioned in or on the at least one first layer of molten material. In step **210**, the at least one second layer of molten material may be deposited on top of the at least one first layer of molten material and the metal plate **24** to encapsulate the metal plate **24** therein. In step **214**, the cavity **20** formed in the body **17** of the second furniture component **14** may be aligned with the metal plate **24** so that the metal plate **24** is aligned with the first furniture component **12** in step **216** when the first furniture component **12** is inserted into the cavity **20**.

[0045] For recycling and/or removal purposes of the first furniture component **12**, the metal plate **24** is heated, for example, via induction. As an example, an induction coil may be placed near the multi-component structure **10** to heat the metal plate **24**. The metal plate **24** may be inductively and separately heated above the melt temperature of the surrounding layers of solidified material of the second furniture component **14** to cause the heat to dissipate to the surrounding layers of material so that the surrounding layers become molten. The metal plate **24** and the first furniture component **12** may then be removed from the second furniture component **14**, either manually with a tool or via gravity acting on the metal plate **24**, the first furniture component **12**, and the surrounding layers of molten material. Alternatively, a resistive heater may be encapsulated in the second furniture component **14** instead of the metal plate **24** and may be used in a similar manner as the metal plate **24** to remove the first furniture component **12**.

[0046] Another method **200'** of three-dimensional (3D) printing a multi-component structure **10'** is shown in FIG. 4. The multi-component structure **10'** is similar to the multi-component structure **10** except for a second side **18'** of a second furniture component **14'**, and the method **200'** is similar to the method **200** except for steps **218'** and **220'**. In step **202**, the first furniture component **12** is provided. In step **204**, the print path for the nozzle to deposit the at least one first layer of molten material is set. In step **206**, the at least one first layer of molten material is deposited. In step **208**, the print path for the nozzle to deposit the at least one second layer of molten material on the at least one first layer is set. In step **210**, the at least one second layer of molten material is deposited on the at least one first layer of molten material while the at least one first layer of molten material is molten.

[0047] In step **212**, the print path for the nozzle to deposit the at least one third layer of molten material on the at least one second layer is set, as shown in FIG. 4. In step **214**, the at least one third layer of molten material is deposited on the at least one second layer of molten material. The print path of the at least one third layer of molten material forms the cavity **20** in the body **17** of the second furniture component **14'** as the at least one second layer of molten material is not covered entirely by the at least one third layer of molten material.

[0048] In step **216**, the first furniture component **12** is inserted into the cavity **20** of the second furniture component **14'**, as shown in FIG. 4. In step **218'**, a print path for the nozzle to deposit at least one fourth layer of molten material is set. The print path is set so that the at least one fourth layer of molten material is only deposited on a perimeter or a portion of the first furniture component **12**. In step **220'**, the at least one fourth layer of molten material is deposited onto the at least one third layer of molten material and the portion of the first furniture component **12** while the at least one third layer of molten material is molten to form the second side **18'** of the second furniture component **14'**.

[0049] The at least one fourth layer of molten material bonds to the at least one third layer of molten material to hold the first furniture component **12** in place, as suggested in FIG. **4**. In some embodiments, the at least one fourth layer of molten material only covers the perimeter or a portion of the first furniture component **12** such that the first furniture component **12** is captured in the second furniture component **14'** by the at least one fourth layer of molten material. In such an embodiment, a portion of the first furniture component **12** is visible and/or accessible from an exterior of the second furniture component **14'**. In step **222**, the layers of molten material are cooled to cause the layers of molten material to solidify and form the second furniture component **14'** having the first furniture component **12** captured therein.

[0050] The methods disclosed herein may also be used to at least partially encapsulate the first furniture component **12** after the second furniture component **14** has been 3D printed. For example, the second furniture component **14** may be retrofit to include the first furniture component **12**. To do so, material from the second side **18** of the second furniture component **14** is removed to form the cavity **20**. In some embodiments, removing the material from the second side **18** of the second furniture component **14** includes milling the material from the second side **18** of the second furniture component **14**. The first furniture component **12** is inserted into the cavity **20** of the second furniture component **14**. Then, at least one layer of molten material is deposited on top of the second side **18** of the second furniture component **14** and the first furniture component **12**. The at least one layer of molten material may be deposited on an entirety of the first furniture component **12** to encapsulate the first furniture component **12**, or the at least one layer of molten material may be deposited on a perimeter or a portion of the first furniture component **12** to capture the first furniture component **12**. Such a method may be used to retrofit 3D printed second furniture components **14** created without the first furniture component **12**. Additionally, such a method may also be used if the first furniture component **12** may not be easily inserted in and/or placed on the second furniture component **14** during the printing process. For example, if the first furniture component **12** is large, vulnerable to heat, and/or not sufficiently stiff, retrofitting the second furniture component **14** to include the first furniture component **12** may be useful.

[0051] Another method **200''** of three-dimensional (3D) printing a multi-component structure **10''** is shown in FIG. **5**. Top-down views of the multi-component structure **10''** and sectional views of the multi-component structure **10''** during the steps of the method **200''** are shown in FIG. **5**. In step **202''**, a first furniture component **12''** is provided. Illustratively, the first furniture component **12''** is glass for a window frame (i.e., a second furniture component **14''**). In step **204''**, a print path for a nozzle of a printer to deposit at least one first layer of molten material on a print surface is set. In step **206''**, the at least one first layer of molten material is deposited on the print surface. The at least one first layer of molten material forms a first side **16''** of the second furniture component **14''**. Steps **204''** and **206''** may be repeated for as many layers as necessary to form a body **17''** of the second furniture component **14''**.

[0052] In step **216''**, the first furniture component **12''** is positioned on the body **17''** of the second furniture component **14''**, as shown in FIG. **5**. In some embodiments, the first furniture component **12''** is positioned on the body **17''** of the second furniture component **14''** by placing the first furniture component **12''** on top of the body **17''**. In other embodiments, the first furniture component **12''** is positioned in a cavity formed in the body **17''** of the second furniture component **14''**, as previously described in relation to methods **200**, **200'**. In step **218''**, a print path for the nozzle to deposit at least one second layer of molten material is set. The print path is set so that the at least one second layer of molten material is deposited on a perimeter of the first furniture component **12''**.

[0053] In step **220''**, the at least one second layer of molten material is deposited onto the at least one first layer of molten material and the first furniture component **12''** while the at least one first layer of molten material is molten to form a second side **18''** of the second furniture component **14''**. The at least one second layer of molten material bonds to the at least one first layer of molten

material to hold the first furniture component **12**" in place. The at least one second layer of molten material covers the perimeter of the first furniture component **12**" such that the first furniture component **12**" is captured in the second furniture component **14**" by the at least one second layer of molten material. In step **222**", the layers of molten material are cooled to cause the layers of molten material to solidify and form the second furniture component **14**" having the first furniture component **12**" captured therein.

[0054] Another method **400** of three-dimensional (3D) printing a multi-component structure **310** is shown in FIG. **6**. Top-down views of the multi-component structure **310** and sectional views of the multi-component structure **310** during the steps of the method **400** are shown in FIG. **6**. In step **402**, a first furniture component **312** is provided. The first furniture component **312** may comprise a placard or a sign, as shown in FIG. **6**, a hook, or a holder, among other items. In some embodiments, the first furniture component **312** includes an external portion **325** and an embedded portion **327**, as shown in FIG. **6**.

[0055] The external portion **325** is configured to be external to a second furniture component **314**, as shown in FIG. **6**. In some embodiments, a surface of the external portion **325** facing the second furniture component **314** is formed to include ridges **326**. The ridges **326** are configured to contact the second furniture component **314** and help to align the first furniture component **312** with the 3D printed second furniture component **314** during the printing of the second furniture component **314**, as suggested in FIG. **6**. In some embodiments, the ridges **326** are milled on the first furniture component **312**.

[0056] The embedded portion **327** of the first furniture component **312** is configured to be embedded in the second furniture component **314**, as shown in FIG. **6**. In some embodiments, the embedded portion **327** is formed to include holes **328**. The holes **328** help to increase a strength of the multi-component structure **310**. Though shown as including a plurality of circular holes **328**, the first furniture component **312** may be formed to include any shape or number of holes.

[0057] In step **404**, a print path for the nozzle of the printer to deposit at least one first layer of molten material on the print surface is set. In step **406**, the at least one first layer of molten material is deposited on the print surface. The at least one first layer of molten material forms a first side **316** of a second furniture component **314**. In step **408**, a print path for the nozzle to deposit at least one second layer of molten material on the at least one first layer is set. In step **410**, the at least one second layer of molten material is deposited on the at least one first layer of molten material while the at least one first layer of molten material is molten. The at least one second layer of molten material forms a portion of a body **317** of the second furniture component **314**.

[0058] In step **412**, a print path for the nozzle to deposit at least one third layer of molten material on the at least one second layer is set. In step **414**, the at least one third layer of molten material is deposited on the at least one second layer. The at least one third layer of molten material forms another portion of the body **317** of the second furniture component **314**. Steps **412** and **414** may be omitted or steps **412** and **414** may be repeated depending on the number of layers needed to form the body **317** of the second furniture component **314**.

[0059] In step **416**, the first furniture component **312** is positioned on the body **317** of the second furniture component **314**, as shown in FIG. **6**. The embedded portion **327** is positioned on top of the body **317** of the second furniture component **314**, and the external portion **325** is positioned adjacent and contacting the second furniture component **314**. The ridges **326** of the external portion **325** of the first furniture component **312** are aligned with each of the layers of molten material.

[0060] In step **418**, a print path for the nozzle to deposit at least one fourth layer of molten material is set. In step **420**, the at least one fourth layer of molten material is deposited onto the at least one third layer of molten material and the first furniture component **312** while the at least one third layer of molten material is molten to form a second side **318** of the second furniture component **314**. The at least one fourth layer of molten material bonds to the at least one third layer of molten material to hold the first furniture component **312** in place. The at least one fourth layer of molten

material may flow through the holes **328** formed in the embedded portion **327** of the first furniture component **312** to bond to the at least one third layer of molten material below the embedded portion **327**.

[0061] In step **422**, the layers of molten material are cooled to cause the layers of molten material to solidify and form the second furniture component **314** having the first furniture component **312** captured and/or partially embedded therein, as shown in FIG. **6**. Method **400** may be advantageously used to partially embed, for example, a coat hook in a chair, a table, or an office pod as a portion of the coat hook may be located external to the second furniture component **314**. [0062] While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

Claims

1. A method of three-dimensional (3D) printing multi-component furniture, the method comprising providing a first furniture component, depositing at least one first layer of molten material on a print surface to form a first side of a second furniture component, depositing at least one second layer of molten material onto the at least one first layer of molten material while the at least one first layer of molten material is molten to form a body of the second furniture component having a cavity, inserting the first furniture component into the cavity of the second furniture component, depositing at least one third layer of molten material onto the at least one second layer of molten material and the first furniture component while the at least one second layer of molten material is molten to form a second side of the second furniture component, and cooling the at least one first layer of molten material, the at least one second layer of molten material, and the at least one third layer of molten material to cause the at least one first layer of molten material, the at least one second layer of molten material, and the at least one third layer of molten material to solidify and form the second furniture component having the first furniture component at least partially encapsulated therein.
2. The method of claim 1, wherein depositing at least one third layer of molten material onto the at least one second layer of molten material and the first furniture component includes depositing the at least one third layer of molten material on a perimeter of the first furniture component so that the first furniture component is captured in the second furniture component by the at least one third layer of molten material and at least a portion of the first furniture component is accessible from an exterior of the second furniture component.
3. The method of claim 1, wherein depositing at least one third layer of molten material onto the at least one second layer of molten material and the first furniture component includes depositing the at least one third layer of molten material on an entirety of the first furniture component so that the first furniture component is entirely encapsulated in the second furniture component by the at least one third layer of molten material.
4. The method of claim 3, wherein the first furniture component is not accessible from an exterior of the second furniture component.
5. The method of claim 3, further comprising, after the step of cooling, removing material from the second side of the second furniture component located above the first furniture component to form an opening in the second side of the second furniture component so that the first furniture component is accessible from an exterior of the second furniture component.
6. The method of claim 5, wherein removing material from the second side of the second furniture component includes milling the material from the second side of the second furniture component.
7. The method of claim 1, further comprising providing a metal plate configured to be embedded or captured in the second furniture component.

- 8.** The method of claim 7, further comprising, after the step of depositing at least one first layer of molten material on a print surface and before the step of depositing at least one second layer of molten material onto the at least one first layer of molten material, locating the metal plate on the at least one first layer of molten material and depositing at least one fourth layer of molten material on the at least one first layer of molten material and the metal plate.
 - 9.** The method of claim 8, further comprising, after the step of cooling, heating the metal plate to cause the solidified at least one third layer of molten material and the solidified at least one fourth layer of molten material located on the metal plate to become molten.
 - 10.** The method of claim 9, further comprising removing the metal plate and the first furniture component from the second furniture component.
 - 11.** The method of claim 1, wherein the first furniture component includes an RFID chip or a NFC chip.
 - 12.** The method of claim 1, wherein the first furniture component includes a sensor.
 - 13.** The method of claim 1, further comprising 3D printing the first furniture component.
 - 14.** A method of three-dimensional (3D) printing a multi-component structure, the method comprising providing a first component, depositing at least one first layer of molten material to form a first side of a second component, depositing at least one second layer of molten material onto the at least one first layer of molten material while the at least one first layer of molten material is molten to form a body of the second component, positioning the first component on the second component, depositing at least one third layer of molten material onto the first component while the at least one second layer of molten material is molten to form a second side of the second component, and cooling the at least one first layer of molten material, the at least one second layer of molten material, and the at least one third layer of molten material to cause the at least one first layer of molten material, the at least one second layer of molten material, and the at least one third layer of molten material to solidify and form the second component having the first component at least partially embedded therein.
 - 15.** The method of claim 14, wherein positioning the first component on the second component includes positioning an embedded portion of the first component on top of the body of the second component and positioning an external portion of the first component adjacent to the body of the second component.
 - 16.** The method of claim 15, wherein depositing at least one third layer of molten material onto the first component includes directing the at least one third layer of molten material through at least one hole formed in the embedded portion of the first component.
 - 17.** The method of claim 15, wherein the external portion of the first component is formed to include a plurality of ridges that align with the first layer of molten material and the second layer of molten material.
 - 18.** A three-dimensional (3D) printed multi-component structure, the multi-component structure comprising a first component, a second component having a first side and a second side opposite the first side, the second component being formed to define a cavity that extends into the second component from the second side toward the first side, wherein the second component is 3D printed and the first component is located in the cavity of the second component to at least partially embed the first component in the second component.
 - 19.** The multi-component structure of claim 18, wherein the first component is entirely encapsulated in the second component.
 - 20.** The multi-component structure of claim 18, wherein the first component comprises a RFID chip, a NFC chip, or a sensor.
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